

CSA0404 – OPERATING SYSTEMS

LAB PROGRAMS

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EXPERIMENT 21:

Question:

Develop a C program to implement the worst fit algorithm of memory management.

Program:



```
1 #include <stdio.h>
2
3 #define MAX 1000
4
5 int memory[MAX];
6
7 void initialize()
8 {
9     for(int i = 0; i < MAX; i++)
10         memory[i] = -1;
11 }
12
13 void display()
14 {
15     int i, start, count;
16
17     printf("\nMemory Status:\n");
18
19     for(i = 0; i < MAX; i++)
20     {
21         if(memory[i] == -1)
22         {
23             start = i;
24             count = 0;
25
26             while(i < MAX && memory[i] == -1)
27             {
28                 count++;
29                 i++;
30             }
31
32             printf("Free Block: %d - %d (%d units)\n", start, i, count);
```

Process 1 allocated from 0 to 199
Process 2 allocated from 200 to 499
Process 3 allocated from 500 to 899

Memory Status:
Free Block: 900 - 999 (100 units)

...Program finished with exit code 0
Press ENTER to exit console.